

1-A:

Asymmetric information refers to a situation where one party has more or better information than the other party in a transaction. In the case of dentistry, the dentist may have more knowledge and information about the patient's teeth and treatment options than the patient does. **This can lead to a power imbalance, where the dentist may recommend unnecessary treatments because the patient is not fully informed or able to make an informed decision.** This explains why key A is correct.

B) Dynamic efficiency refers to the ability of a market to allocate resources efficiently over time.

C) Positive externalities refer to benefits that accrue to third parties, not directly involved in a transaction.

D) Social benefits refer to benefits that accrue to society as a whole, which include private benefits and external benefits.

2-B:

Key take-away: The rational consumer is indifferent along the indifference curve. This is because all consumption bundles on the indifference curve yield the same level of total utility.

An indifference curve represents different combinations of goods (apples and pears in this case) that provide the same level of total utility or satisfaction to the individual. Since point X and point Y lie on the same indifference curve (I1), it means that the individual derives the same level of total utility from both combinations.

The movement from point X to point Y represents a substitution of apples for pears, but the overall utility remains constant.

A) This may be because the individual can afford more pears than apples, because apple's relative prices may have risen.

C) This represents that the individual now prefers pears to apples.

D) Since the individual remains on the same indifference curve, their total utility does not increase or decrease, it remains the same.

3-D:

Price discrimination requires:

1. Ability to separate markets (A)
2. Different price elasticities of demand in each market (B)
3. Monopoly power (C) - the ability to control prices in each market

Price leadership (D) is not necessary for price discrimination. Price leadership refers to a situation where one firm sets the price and other firms follow.

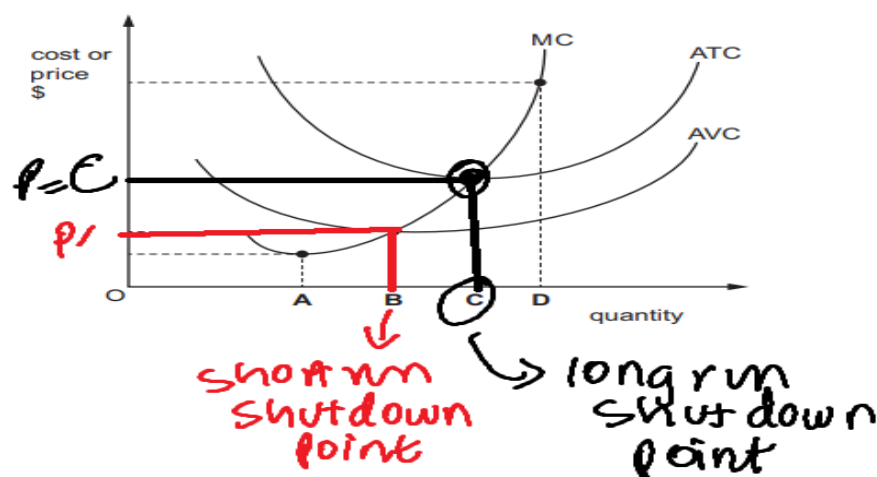
Price discrimination occurs when a firm charges different prices for the same product or service in different markets, exploiting differences in price elasticity of demand. **The firm must have some degree of market power to implement price discrimination, but price leadership is not a necessary condition.**

This explains why key D will be correct.

4-C:

The diagram shows the cost curves of a firm.

What is the level of output below which the firm will shut down in the long run?



In the long-run, a firm shuts down, if it fails to cover its total costs, or in other words, when $AR < ATC$.

This explains why key C will be correct as below point C (where $AR = ATC$), the firm would not continue to operate in the long-run.

5-C:

By taking over its main competitor, Steel producer X would expect:

- Average Revenue (**price**) to increase: With a larger market share, X can exert greater market power, potentially leading to **higher prices** and increased revenue.
- Long-run Average Cost to decrease: With the takeover, X can eliminate redundant costs, consolidate operations, and take advantage of economies of scale, leading to **lower average costs in the long run**.

This explains why key C will be correct.

6-C: The correct equation is:

Social benefit = Private benefit + External benefit

Rearranging this equation, we get:

Private benefit = Social benefit - External benefit

Therefore, option C is the correct statement.

Here's a brief explanation:

- Private benefit refers to the benefits enjoyed by an individual or firm.
- External benefit refers to the benefits enjoyed by third parties, not directly involved in the transaction
- Social benefit refers to the total benefit, including both private and external benefits.

7-B: Moral hazard occurs when an individual or organization takes on more risk than they would normally because they are insulated from the consequences of their actions (e.g., by insurance).

To overcome moral hazard, insurance contracts can be designed to share the risk between the insurance company and the client, such as:

- The client pays a portion of the loss
- The client pays a portion of the costs

This way, the client has some "skin in the game" and is incentivized to take less risk, reducing moral hazard.

A) Charging the same premium to all people in the same age group can actually create moral hazard, as low-risk individuals may be over-insured.

C) Unconditional financial support to banks can create moral hazard, as banks may take on excessive risk knowing they'll be bailed out.

D) Guaranteeing jobs for bank managers can create moral hazard, as they may take on excessive risk knowing they'll keep their job regardless.

8-C:

A – If a child is vaccinated against measles, which prevents other children from catching the disease, this will represent **positive externality of consumption**.

B – If a consumer donates products to a local food bank, this would incur cost to the individual who is donating.

C - A **positive externality** of production occurs when a firm's production process generates a benefit that falls on third parties, such as society as a whole. In this case, the firm's action of rebuilding the local road benefits not only the firm but also the community, making it easier for people to commute, transport goods, and access services. This explains why it will be regarded as a positive production externality.

D- When a government pays subsidies to firms producing cars, this will represent an internal cost borne by the government.

9-B:

- **Productive efficiency** = No: The firm's lack of competition and high barriers to entry/exit lead to X-inefficiency, meaning it doesn't minimize its production costs. It may not adopt cost-saving technologies or optimize its production process. Key take-away: **X inefficiency is when there is no incentive to work hard due to a lack of competitive pressures.**

- **Allocative efficiency** = No: A monopoly produces where $MC=MR$ i.e. the profit maximizing level of output which is lesser than the socially desirable level of output where $MC=AR$ or $MSB=MSC$. This results in a deadweight loss, as the firm prioritizes its own profits over social welfare.
- **Dynamic efficiency** = Yes: The firm's position (that enables it to earn super-normal profits) encourages innovation and investment in research and development (R&D) to maintain its market advantage and protect its barriers to entry. This leads to technological progress and potential future productivity gains.

In this case, the firm's market structure and position lead to inefficiencies in the short run (productive and allocative), but incentivize innovation and investment in the long run (dynamic efficiency)

10-B:

By increasing patent protection, the government makes it more difficult for new firms to enter the industry by **increasing “legal barriers”**

- Strengthening intellectual property rights: Inventors and existing firms have more legal protection, making it harder for new entrants to access technologies and knowledge.

Barrier to Entry	Definition	Example
Cost Barriers	Costs that new firms must incur to enter an industry, making it difficult for them to compete with established firms.	High startup costs, advertising expenses, research and development costs.
Market Barriers	Factors that make it difficult for new firms to gain market share, such as loyal customers, established brands, and network effects.	Strong brand recognition, customer loyalty programs, exclusive contracts.
Physical Barriers	Physical constraints that limit access to resources, markets, or customers, making it hard for new firms to enter.	Limited access to raw materials, geographical constraints, transportation costs.

This explains why only key B is correct.

Patents are a legal construct, not a natural or market-driven constraint. Patents are a form of intellectual property protection that can be acquired and used to exclude others from producing or selling a particular invention. While patents can create a barrier to entry, they are a legal barrier.

11-A:

It can be seen that when the third labor is employed, the ACL and the MCL fall.

Workers	Total Cost (TC)	Average Cost (AC) <i>of labor</i>	Marginal Cost (MC) <i>of labor</i>
1	\$1200	\$1200	-
2	\$2000	\$1000	\$800 (\$2000 - \$1200)
3	\$2700	\$900	\$700 (\$2700 - \$2000)

This explains why key A will be correct.

12-D:

Cost-benefit analysis is a method used to evaluate projects by comparing their SB and SC usually in monetary terms. However, national pride is a non-monetary, intangible benefit that is difficult to quantify and value precisely. While it may be an important benefit, it is challenging to assign a specific dollar value to it, making it hard to include in a cost-benefit analysis. **This explains why key D will be correct.**

A) Importing materials may affect the cost of the project, but it doesn't make it difficult to use cost-benefit analysis.

B) Here we are able to calculate the external costs such as that associated with clearing land, and compare it with private costs as a result of which, it can be said that **the external costs of clearing land will outweigh the private costs**. This will ultimately help us while performing the cost benefit analysis.

C) The potential underutilization of the stadium **after the World Games** is not a relevant consideration.

13-C:

$$\text{MRP} = \text{MPP} \times \text{MR}$$

- MRP = Marginal Revenue Product
- MPP = Marginal Physical Product (additional output produced by one more worker)
- MR = Marginal Revenue (change in total revenue resulting from the sale of one more unit of output)

So, if MPP (Marginal Physical Product) increases, MRP also increases (assuming constant MR), leading to a rightward shift of the demand curve for labor. This explains why key C will be correct.

A) A decrease in marginal revenue productivity would lead to a leftward shift of the demand curve, not rightward.

B) A decrease in the wage rate would lead to an increase in the quantity of labor demanded, but it would not shift the demand curve.

D) An increase in the wage rate would lead to a decrease in the quantity of labor demanded, not a shift in the demand curve.

14-D:

Employment	Total Product (TP)	Marginal Product (MP) <i>→ change in TP</i>	Average Product (AP) <i>→ TP/L</i>
0	0	-	-
1	6.0	6.0 (6-0)	6.0 (6/1)
2	13.0	7.0 (13-6)	6.5 (13/2)
3	21.0	8.0 (21-13)	7.0 (21/3) <i>→</i>
4	27.5	6.5 (27.5-21)	6.88 (27.5/4)
5	33.5	6.0 (33.5-27.5)	6.7 (33.5/5)

This explains why key D will be correct, as when the API is 7 at employment level of 3, MPL is 8.0.

15-B:

- Education and training can help increase the productivity and skills of workers, especially from lower-income backgrounds.
- This can lead to better job opportunities and higher wages, reducing income inequality.
- Government spending on education and training can also help level the playing field, providing more equal access to opportunities for all. **This explains why key B will be correct.**

A) Reducing the top tax rate would likely increase income inequality, as the wealthy would benefit more from tax cuts.

C) Increased unemployment would likely worsen income inequality, as those without jobs would struggle to make ends meet.

This is because individuals will be more likely to be trapped into the vicious cycle of poverty, as can be seen below.

1. Unemployment → Reduced income → Poverty
2. Poverty → Limited access to education & training → Reduced skills & productivity
3. Reduced skills & productivity → Limited job opportunities → Unemployment
4. Unemployment → Reduced income → Poverty (back to step 1)

D) Privatizing the education sector could lead to unequal access to quality education, exacerbating income inequality.

16-C:

The correct answer is indeed C) Certainty of employment for skilled immigrants was made the main criterion.

In the new Express Entry system, the "job offer" criterion has been introduced and heavily weighted indicating that having a job offer in Canada is now a primary consideration for skilled immigrants. This suggests that Canada prioritizes candidates who already have a job lined up, ensuring they can contribute to the economy immediately.

A) Canada intended to maintain the same level of skilled immigrants under both systems.

- This is incorrect because the weighting changes suggest a shift in priorities, not maintaining the same level.

B) Canada wanted a smaller number of highly educated skilled immigrants.

- This is incorrect because the decrease in the weighting given to education only represents that some factors are regarded more important than another. The criteria weighting in % form will not help us to deduce the number of highly educated skilled immigrants that it wants.

D) Skilled immigrants needed to have fewer years of work experience than previously.

- This is incorrect because while the weighting for work experience decreased, it's still a consideration ; this only represents that some factors are regarded more important than another. The criteria weighting in % form will not help us to deduce the number of years of experience required.

17-C:

Imposing a maximum price for electricity can be justified on economic grounds as a way to prevent monopolistic exploitation, where a single supplier or a small group of suppliers might otherwise charge high prices, taking advantage of their market power. This is how the government regulates monopolies. **As a result of which, key C, becomes correct.**

A) Imposing a maximum price might actually discourage investment in additional capacity, as suppliers contract quantity supplied when a price cap legally reduces prices that they can charge.

B) Since a lower price would result in an extension in quantity demanded of electricity, this represents that there will be no incentive for consumers to conserve energy.

D) A price cap could further lead to shortages since QD extends, while QS contracts, mandating the use of rationing mechanisms to allocate scarce resources among consumers, most probably on the basis of a first come, first served basis.

18-A:

The correct answer is A

A) It is spending on capital goods by firms because the rate of growth in demand for final goods and services increases.

-> Demand-induced investment refers to the increase in investment by firms in response to an increase in demand for their products. When the rate of growth in demand grows, firms invest in new capital goods, such as machines, equipment, and buildings, to increase production capacity and meet the rising demand. **According to accelerator theory, induced investment is a function of a rate of change in income.**

Consumption-> AD increases-> RNI rises-> investment rises.

B) A fall in the cost of borrowing (interest rates) encourages investment as private producers consider **marginal efficiency of investment**, but it's not the same as demand-induced investment.

C) Investing in new technology to improve productivity is related to improvements in the **supply side of the economy**.

D) Replacement of old capital goods with new ones is related to **autonomous investment (which is independent on national income)**

Key take-away: Induced investment is dependent on national income.

19-D:

A) Reducing debt burdens enable governments to invest in services that support economic development.

B) Financing capital projects that improve access to basic services like clean water is essential for human capital development and economic productivity (financial aid)

C) Creating employment opportunities and transferring skills and knowledge to workers from low-income countries increase their human capital and potential for economic growth.

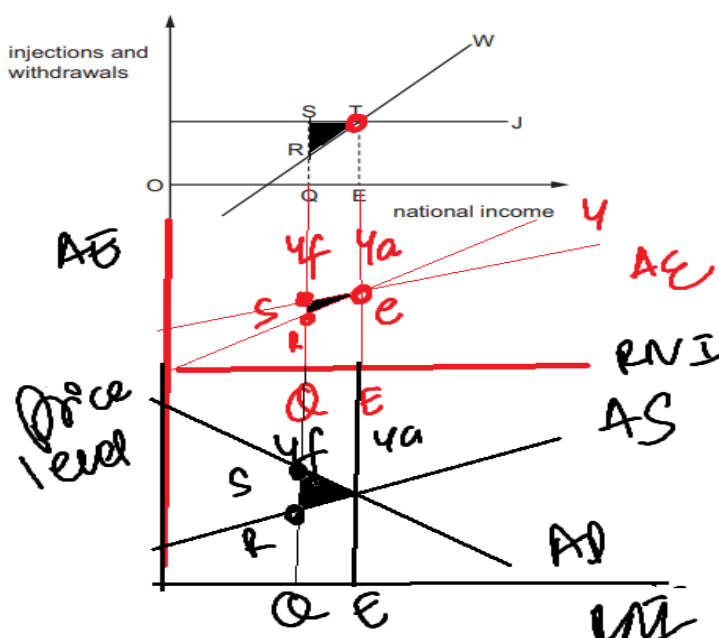
D) Reducing fuel exports to low-income countries would likely harm their economic growth, as many of these countries rely on imported energy to power their economies. This policy would lead to energy shortages, increased costs, and decreased economic activity in the low-income countries.

This explains why key D will be incorrect.

20- C: Inflationary gap can be either of the following.

1. **Excess of Aggregate Demand (AD) over Aggregate Supply (AS):** This definition refers to a situation where the total demand for goods and services in an economy exceeds the total supply. This excess demand leads to an inflationary pressure, as businesses respond to the demand surge by increasing prices.
2. **Excess of Aggregate Expenditure (AE) over Potential Real National Income:** This definition looks at the difference between the total amount spent in the economy and the maximum potential output of the economy . When AE exceeds potential output, it leads to an inflationary gap.
3. **Excess of Injections over Withdrawals:** This definition focuses on the flow of funds into and out of the economy. Injections represent new spending or investment (e.g., government spending, exports, gross investment), while withdrawals represent leakages from the circular flow (e.g., savings, imports, tax). **When injections exceed withdrawals, it leads to an increase in aggregate demand and an inflationary gap.**

An open economy has injections, J, into the circular flow of income and withdrawals, W.



With reference to this question, inflationary gap represents the excess of injections over withdrawals, at full employment level of income i.e Q. In this case, it will be represented through "SR"

21-D:

The national income multiplier in an open economy with a government sector is calculated as:

$$\text{Multiplier} = 1 / (\text{MPS} + \text{MPT} + \text{MPM})$$

A low marginal propensity to import (MPM) means that a smaller portion of the increased income is spent on imports, leaving more income to be spent domestically. This leads to a higher multiplier effect, as the increased spending has a larger impact on domestic income and employment. **This explains why key D will be correct.**

The other options would actually reduce the multiplier effect:

A) High average and marginal tax rates (MPT) would reduce the multiplier, as more income is taken away from households and businesses.

B) High marginal propensity to save (MPS) would also reduce the multiplier, as a rise in withdrawals represents the fact that marginal propensity to consume will fall.

C) Autonomous consumption refers to consumption that is not dependent on income, such as consumption funded by wealth. When autonomous consumption is less, it means that there is a lesser injection of spending into the economy, regardless of income levels.

22-B:

The correct answer is B) frictional.

It needs to be understood that involuntary unemployment is different from voluntary unemployment. Voluntary unemployment refers to a situation when people are able to work, but are not willing to work a certain job although suitable work is available for them. In other words, they are voluntarily unemployed, i.e., unemployed of their own will.

On the contrary, involuntary unemployment occurs when those who are able and willing to work at the going wage rate do not get work. Hence, they are unemployed against their wishes.

Usually, voluntary unemployment is a type of unemployment in which, workers are unable to accept a certain job at the going wage rate, because of a very less difference between unemployment benefits and the take-home pay of the employed. This may be usually because of generous unemployment benefits. This type of unemployment is categorized under the category of “frictional unemployment”. As a result of which, key B, is correct.

23- B:

-> Crowding out occurs when government increases its public spending by borrowing from the non-bank private sector by issuing bonds. This leads to reduced deposits in commercial banks as individuals now purchase securities. When this leads to a decrease in the supply of loanable funds, this results in higher interest rates, which in turn reduce private investment expenditure by increasing their costs of borrowing.

This explains why key B becomes correct.

A- Automatic stabilizers are government programs that automatically increase or decrease spending in response to changes in economic activity, without the need for explicit policy changes.

C- The accelerator effect examines the effect on levels of induced investment from a change in income.

D- The substitution effect is a phenomenon in which changes in relative prices lead to changes in consumer behavior, as consumers substitute one good for another.

24-A : If all economic policy aims are achieved, it means that the government has successfully implemented policies to:

- Promote economic growth
- Control inflation
- Reduce unemployment
- Improve the balance of payments.

Achieving these aims would likely lead to:

- Increased economic growth, which would raise the average standard of living
- Low and stable inflation, which would maintain purchasing power
- Reduced unemployment, which would increase employment opportunities
- Improved trade balance, which would strengthen the currency's foreign exchange rate.

Note that all of these objectives result in improvement of living standards.

Therefore, a decrease in the average standard of living (option A) would be the least likely outcome if all economic policy aims are achieved.

This explains that only key D can be correct.

25-A:

A) Raising the rate of value-added tax (VAT) is a fiscal policy measure that increases government revenue. This suggests that the government's main macroeconomic objective is to reduce the budget deficit, which is the difference between government spending and revenue. **This explains why key A will be correct.**

B) VAT (which is an indirect tax) is a regressive tax, meaning it disproportionately affects lower-income households, who spend a larger portion of their income on consumption. Therefore, raising VAT may actually exacerbate income inequality, rather than reduce it.

C) Raising VAT is not a direct measure to reduce unemployment. In fact, higher VAT rates may lead to higher prices and reduced real incomes which result in reduced consumption, and potentially lower demand for labor, which could even increase demand-deficient unemployment.

D) Indirect taxes are a tax on goods and services, which raise the costs of production of firms, as a result of which, when Aggregate Supply reduces, there is an increase in cost push inflation.

26-D:

When the economy's exchange rate depreciates:

- **Key take-away: exported prices only decrease in event of depreciation when we are talking about foreign currency terms.**
- The price of imported goods in the domestic currency will rise (because the weakened currency can buy fewer units of foreign goods)
- The price of exported goods in the domestic currency will also rise (because the exported goods are now more expensive to produce due to the increased cost of imported inputs. **This will be because, we are asked about the impact on the prices of exported goods in the “domestic currency”**)

27-D: The Human Development Index (HDI) is a composite measure of three dimensions:

1. Life expectancy at birth
2. Mean years of schooling and expected years of schooling (education)
3. Real national income per capita (GNI per capita)

An increase in the HDI value from 0.60 in 2014 to 0.64 in 2020 indicates an overall improvement in human development in India. Since life expectancy and real national income per capita are unlikely to decrease (given the overall increase in HDI), the most likely conclusion is that there was an increase in the mean years of schooling of the adult population, **which makes option D correct.**

A) Life expectancy at birth is likely to have increased given the rise in HDI.

B) Real national income per head is likely to have increased given the rise in HDI

C) Environmental pollution and crime rates are not directly related to the HDI index. Instead, they are adjusted when calculating measure of economic welfare.

28-A:

The J-curve effect refers to the initial worsening of the current account balance after a depreciation of the exchange rate, followed by an improvement. This is due to a time lag in adjusting to the new prices. These adjustments take time, causing the initial deterioration of the current account balance before it improves. **As a result of which, key A, becomes correct.**

C and D: The demand for imports and exports is inelastic in the short-run, and also the supply of goods is inelastic in the short-run.

B: J curve does not have anything to do with the quality of goods produced.

29-C:

The country that is likely to be the most **highly developed** would have:

- A low Gini coefficient (indicating low income inequality)
- A low death rate (indicating good healthcare and living standards)
- A high relative size of the tertiary sector (indicating a strong service sector and a high level of economic development)
- This is why, key C, becomes correct.

30-B:

B) Economic growth refers to an increase in a country's production capacity, often measured by the increase in Gross Domestic Product (GDP). While economic growth is essential for economic development, the two concepts are not identical. This explains why key B is correct.

A) An expansion of economic choices is a key aspect of economic development.

C) Eradication of absolute poverty is a key goal of economic development.

D) Satisfaction of basic needs is a fundamental aspect of economic development.

